

Understanding MLOps

The term MLOps is coined from the words 'machine learning' and 'operations'. It is a practice for collaboration and communication between data scientists and operations professionals to help manage the production machine learning (or deep learning) life cycle. It is similar in concept, though not the same as DevOps.



Let's delve into this emerging world of Machine Learning Ops (MLOps) to understand its various facets. Machine learning (ML) is all pervasive nowadays. If we look at the various categorisations of the ML investment spectrum, ML applications take the largest pie of \$US 28 billion, whereas ML platforms follow with US\$ 14 billion. These two comprise more than 80 per cent of the total investments. Spending on machine learning is estimated to reach US\$ 57.6 billion dollars by 2021, at a compound annual growth rate (CAGR) of 50.1 per cent.

With this kind of proliferation, there is bound to be a plethora of tools available at our disposal to implement an end-to-end ML workflow. This is where the emerging trends of Machine Learning Ops (MLOps) come into

play. MLOps has some similarities with DevOps, but one of the major differences between the two is that MLOps is about managing the deployment of ML models and pipelines, whereas DevOps is about the deployment of the code. In this article, we will investigate the taxonomy of a few MLOps tools.

The MLOps concept

Let us look at the sample life cycle of MLOps, courtesy NVIDIA.

The overall idea of MLOps is simple yet intriguing. There are two ways to look at it — first, one can consider it as a connected flow of various data-centric operations to explore, manage, process and visualise the data to get the model out of it. This is shown in Figure 1. It shows a continuum of data extraction (data needs) with data marshalling

(data fixes) to make it appropriate for ML modelling. The ML part of the workflow consists of the usual training, validation, testing and deployment. The loop continues with successive training to adopt the changes in data and hence the underlying model.

Second, MLOps is a layer sitting on top of the usual DevOps. Basically, it is a logical extension of DevOps. This is more relevant when one talks about a logical migration from existing DevOps to MLOps. Figure 2 depicts the same.

The technical debt metaphor

In a paper titled 'Hidden Technical Debt in Machine Learning Systems' written by a few distinguished Google ML practitioners, some pertinent questions have been raised. These are the fundamental questions that MLOps seeks an answer to. The basic idea is simple. How can one take a look

